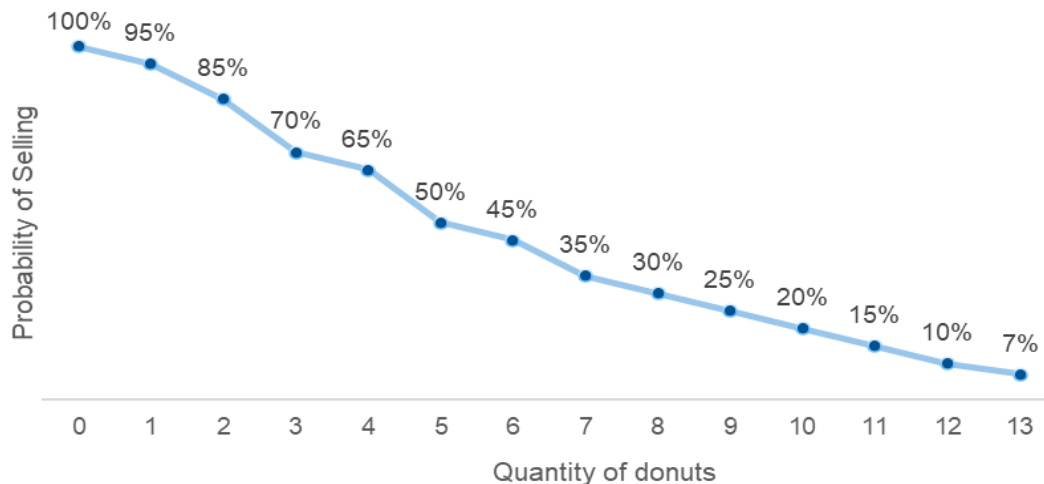


News vendor Model (Answer Key)

Task 1:

A coffee shop sells their special donuts from 8 AM to 6 PM. It costs the shop 0.5 € to buy the donuts from a manufacturer, and they sell them for 2 € each. The donuts last exactly one day and have 0 salvage value. Any unsold donut must be disposed of, and it is a loss to the coffee shop. The probability of selling the donut is provided in the graph below and in the adjacent table.



- a. What is the probability of meeting all demands?

Answer: 75%

- b. How many donuts should the coffee shop order?

Answer: 9

Task 2:

A flower vendor in the Netherlands sources bouquets from the floral retailer at the cost of 16.5 € each and sells them at a price of 50€ each. The demand for the bouquet on any given day is assumed to be normally distributed with a mean of 2500 and a standard deviation of 450. The retailer does not take back any unsold bouquets, and the unsold bouquet must be thrown away.

- a. Calculate the optimal service level. How does the service level change if the standard deviation of demand distribution changes for all other parameters being equal?

Answer: 67%; Service level is independent of demand distribution, and it is not affected by a change in standard deviation (σ).

- b. Calculate the optimal order quantity. How does the optimal order quantity change if the standard deviation changes for all other parameters being equal?

Answer: 2698; Optimum order quantity, Q^* , (for $SL > 50\%$) increases if the standard deviation is increased.

- c. How does optimal order quantity change when the Service level changes with standard deviation?

Answer:

- Optimum order quantity, Q^* , (for $SL > 50\%$) increases if the standard deviation is increased.
- Optimum order quantity, Q^* , (for $SL < 50\%$) decreases if the standard deviation is increased.
- Optimum order quantity, Q^* , (for $SL = 50\%$) remains the same as that of mean demand if the standard deviation is changed.

Task 3:

The underage cost for a watch is \$3495, and the overage costs are \$2840. The demand for these watches on any given day can be assumed to be normally distributed with a mean of 20 and a standard deviation of 4.

- a. What is the optimal in-stock probability?

Answer: 55.17%

- b. How many units must be ordered each week, considering the store is open from Monday - Saturday? Also, calculate what is the optimal order quantity when the store is operating only on Sunday.

Answer:

- Optimal order quantity for Sunday = 20
- Optimal order quantity for Monday – Saturday = 121